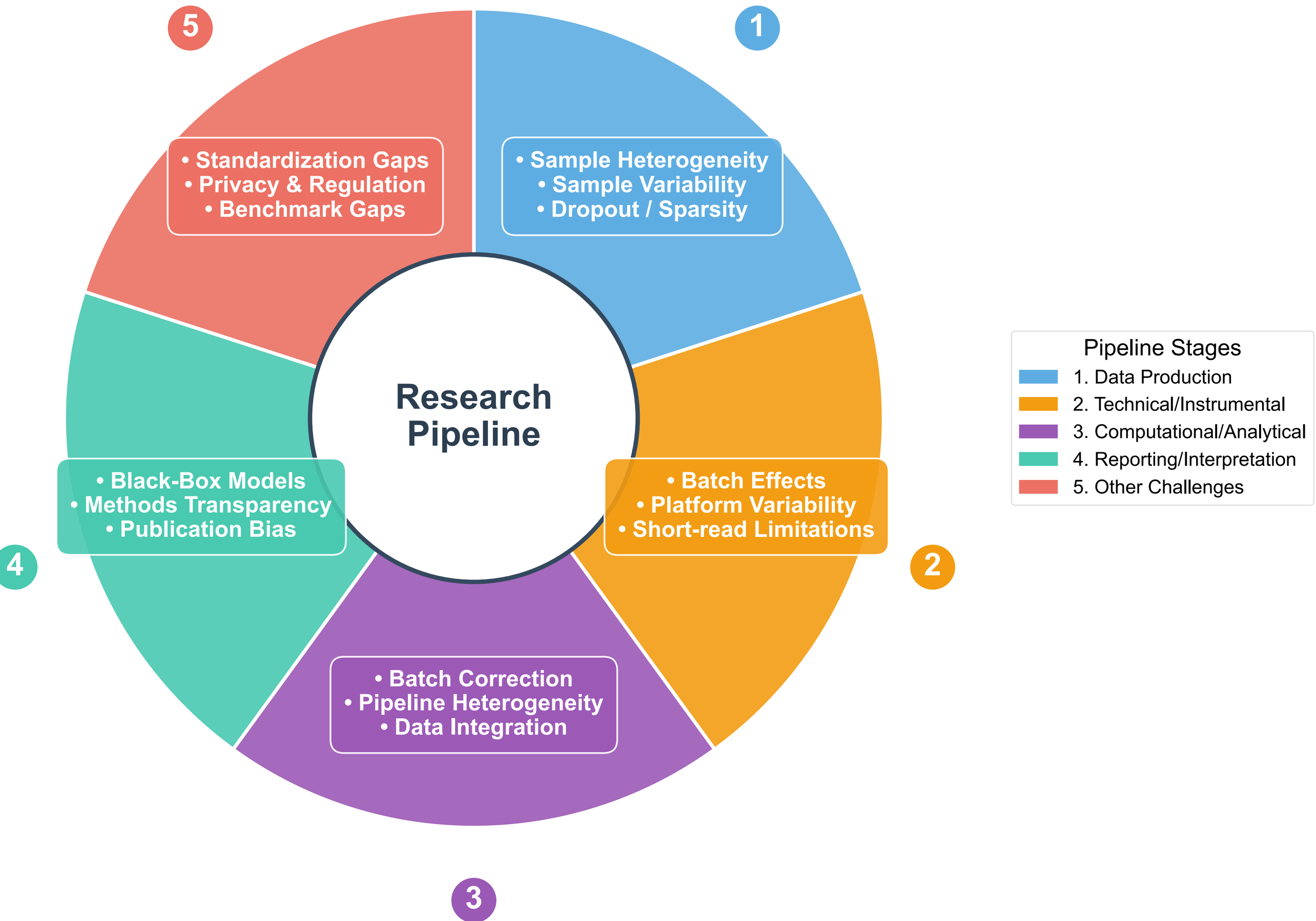
5
1
• Standardization Gaps
• Privacy & Regulation
• Benchmark Gaps
• Sample Heterogeneity
• Sample Variability
• Dropout / Sparsity
Research Pipeline
• Black-Box Models
• Methods Transparency
• Publication Bias
• Batch Effects
• Platform Variability
• Short-read Limitations
4
2
• Batch Correction
• Pipeline Heterogeneity
• Data Integration
3
Pipeline Stages
1. Data Production
2. Technical/Instrumental
3. Computational/Analytical
4. Reporting/Interpretation
5. Other Challenges